

DSA 210 - Final Report

Predicting Football Player Market Values Using Performance Statistics

1. Motivation

Player transfer market values in professional football are often considered subjective, influenced by media hype, agent negotiations, and club budgets. This project investigates whether a player's market value can be objectively explained and predicted using seasonal on-pitch performance statistics. The aim is to bring data-driven objectivity to a process that is often opaque, with potential applications in club recruitment and analytics. The analysis covers Europe's top five leagues (Premier League, La Liga, Bundesliga, Serie A, Ligue 1) for the 2024-25 season.

2. Data Source

Two publicly available datasets were combined. The primary dataset, containing on-pitch performance statistics, was sourced from FBref via a Kaggle compilation. This dataset provides 37 features per player, including goals, assists, expected goals (xG), expected assisted goals (xAG), progressive carries, progressive passes, and defensive actions. The enrichment dataset, providing market valuations, was obtained from the Transfermarkt Kaggle dataset, which aggregates crowdsourced market value estimates for thousands of players worldwide.

The two datasets were merged on normalized player names (lowercase, whitespace removed). Players with fewer than 500 minutes played were filtered out to reduce noise. The final cleaned dataset contained 1,655 players with each player's market value (in euros) as the target variable.

3. Data Analysis

The analysis was conducted in three stages: exploratory data analysis (EDA), hypothesis testing, and machine learning.

Exploratory Data Analysis: The market value distribution was found to be heavily right-skewed, with the median around €7M and a long tail extending past €180M. A log transformation was applied to make the distribution more symmetric for modelling. Position-based and league-based breakdowns, age vs value scatter plots, and correlation analysis were also performed.

Hypothesis Testing: A Shapiro-Wilk test confirmed the market values were not normally distributed, so non-parametric tests were used throughout ($\alpha = 0.05$). Four hypotheses were tested using Kruskal-Wallis, Mann-Whitney U, and Spearman correlation.

Machine Learning: Three regression models were trained to predict $\log(\text{market value})$: Linear Regression as a baseline, Random Forest, and Gradient Boosting. The dataset was split 80% training and 20% testing, with 5-fold cross-validation used for validation.

4. Findings

Position-based analysis showed clear differences: Forwards (median €8.0M) and Midfielders (€8.0M) commanded higher valuations than Defenders (€5.0M) and Goalkeepers (€4.5M). By league, the Premier League consistently had the highest median market values. Correlation analysis identified playing time (90s, Min), goals (Gls), and progressive actions (PrgC, PrgP) as the strongest correlates of market value.

All four null hypotheses were rejected. H1 (Kruskal-Wallis) confirmed that market values differ significantly across positions, and H2 (Kruskal-Wallis) confirmed they differ across leagues. H3 (Mann-Whitney U) showed that Forwards have significantly higher market values than Defenders. H4 (Spearman correlation) confirmed that xG is positively and significantly correlated with market value.

Gradient Boosting outperformed both Random Forest and the Linear Regression baseline in terms of R^2 and RMSE. Feature importance analysis from the Random Forest confirmed that playing time, goal contributions, and progressive actions were the most predictive features. This aligns with the EDA correlation findings and confirms the intuition that consistent, attacking-oriented performance is the strongest predictor of value.

5. Limitations and Future Work

The main limitation of this project is that the model only uses on-pitch performance statistics from a single season. Market values are also influenced by subjective factors such as age trajectory, contract length, marketability, injury history, and club financial situations - none of which were captured here. The dataset was also limited to outfield players in the top five European leagues, so the model may not generalize well to other leagues or age groups.

Future work could enrich the model by including contract length, age trends, social media following (as a proxy for marketability), and historical injury data. Including multiple seasons would allow time-series analysis of value progression. A classification approach predicting market value brackets (e.g. under 5M, 5-20M, over 20M EUR) might also be more useful for practical recruitment applications than a regression on the exact euro value.

6. References

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